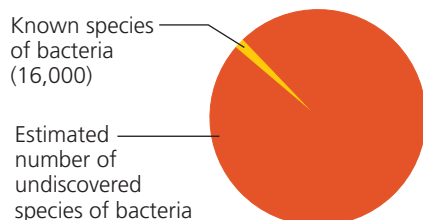


▼ Figure 27.17 Exploring Bacterial Diversity

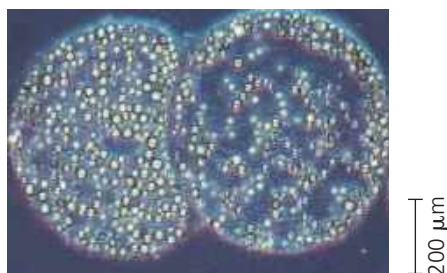
This figure highlights a few key groups of bacteria, but their actual diversity is far greater than shown here. Likewise, gene sequence data indicate that the 16,000 known species of bacteria represent a small fraction of the actual number, estimated at 700,000–1.4 million species.



Proteobacteria

This large and diverse clade of gram-negative bacteria includes photoautotrophs, chemoautotrophs, and heterotrophs. One autotroph, the sulfur bacterium *Thiomargarita namibiensis*, obtains energy by oxidizing H_2S , producing sulfur as a waste product (the small globules in the photograph below). Heterotrophs include pathogens such as *Neisseria gonorrhoeae*, which causes gonorrhea; *Vibrio cholerae*, which causes cholera; and *Helicobacter pylori*, which causes stomach ulcers. Current evidence indicates that mitochondria evolved by endosymbiosis from a heterotroph in the subgroup alpha proteobacteria.

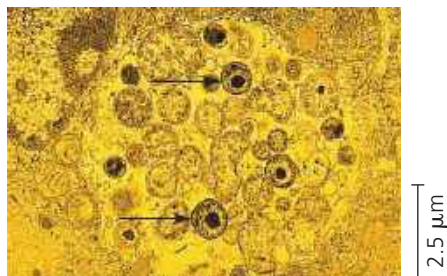
Thiomargarita namibiensis containing sulfur wastes (LM)



Chlamydias

These parasites can survive only within animal cells, depending on their hosts for resources as basic as ATP. The gram-negative walls of chlamydias are unusual in that they lack peptidoglycan. One species, *Chlamydia trachomatis*, is the most common cause of blindness in the world and also causes nongonococcal urethritis, the most common sexually transmitted disease in the United States.

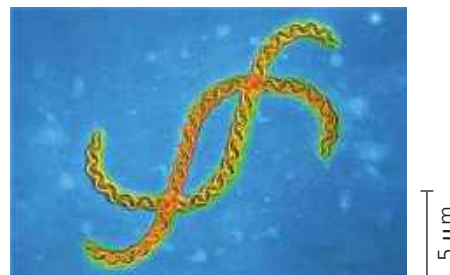
Chlamydia (arrows inside an animal cell (colorized TEM))



Spirochetes

These helical gram-negative heterotrophs spiral through their environment by means of rotating, internal, flagellum-like filaments. Many spirochetes are free-living, but others are notorious pathogenic parasites: *Treponema pallidum* causes syphilis, and *Borrelia burgdorferi* causes Lyme disease.

Leptospira, a spirochete (colorized TEM)



Cyanobacteria

These gram-negative photoautotrophs are the only prokaryotes with plantlike, oxygen-generating photosynthesis. (In fact, chloroplasts are thought to have evolved from an endosymbiotic cyanobacterium.) Both solitary and filamentous cyanobacteria are abundant components of freshwater and marine *phytoplankton*, the collection of small photosynthetic organisms that drift near the water's surface.

Cylindrospermum, a filamentous cyanobacterium



Gram-Positive Bacteria

Gram-positive bacteria rival the proteobacteria in diversity. Species in one subgroup, the actinomycetes, form colonies containing branched chains of cells; two of these species cause tuberculosis and leprosy, but most are decomposers living in soil. Soil-dwelling species in the genus *Streptomyces* are cultured as a source of antibiotics, including tetracycline and erythromycin. Other subgroups of gram-positive bacteria include pathogens such as *Staphylococcus aureus* (see Figure 22.14), *Bacillus anthracis*, which causes anthrax, and *Clostridium botulinum*, which causes botulism.

Streptomyces, the source of many antibiotics (SEM)

